

1 Problem statement

There are 2026 integers greater than 1 written on a blackboard. In a single move two numbers $n > 1$ and $m > 1$ are selected and replaced with

$$a = \gcd(m, n)$$

$$b = \frac{\text{lcm}(m, n)}{\gcd(m, n)}$$

The process continues while it is possible to pick two such numbers. Prove that regardless of the number selection order, after finitely many moves a single one number $M > 1$ will remain on the board, the rest will be equal to 1, and the value of M doesn't depend on the selection choices.

2 Solution draft

First, let's prove that the process will converge. There are $C = 2026$ numbers greater than 1 initially, let's say the product of all 2026 numbers is P . After each move, we have two options:

- (a) If $\gcd(m, n) = 1$, $C := C - 1$
- (b) If $\gcd(m, n) > 1$

$$ab = \gcd(m, n) \frac{\text{lcm}(m, n)}{\gcd(m, n)} = \frac{mn}{\gcd(m, n)}$$

$$P := P / \gcd(m, n)$$

As both C and P are finite, in finite number of steps we'll get to a point where no move is possible. Let's check that no valid move can produce $a = b = 1$. If $a = \gcd(m, n) = 1$, for some $m, n > 1$, $b = \frac{\text{lcm}(m, n)}{\gcd(m, n)} = \text{lcm}(m, n) = mn > 1$. Thus, the process will converge, there's no way for it to end with all numbers equal to 1, so we'll get single number $M > 1$.

Let's denote original numbers as N_i , $1 \leq i \leq 2026$. Let's represent each number as its prime factorization:

$$N_i = \prod_{j=1}^k p_j^{d_{i,j}}$$

where $p_1 < p_2 < \dots < p_k$ are primes and each $d_{i,j} \geq 0$. Let's assume p_k is the largest prime factor among all 2026 numbers.

Let's look at specific factor p_j across all N_i . When two numbers m and n are selected for the next move, let's call the exponents for that prime in m and n factorization d_m and d_n . Then, the exponents of p_j in replacements would be

$$d_a = d_{\gcd} = \min(d_m, d_n),$$

$$d_b = d_{\text{lcm}} - d_{\gcd} = \max(d_m, d_n) - \min(d_m, d_n) = |d_m - d_n|$$

If one or both of d_m or d_n equal 0, $d_a = 0$ and $d_b = \max(d_m, d_n)$, so the exponents don't change.

If not, such construction is equivalent to Euclid algorithm. Euclid algorithm generalizes to $n > 2$ numbers due to transitivity of $\gcd$ operation.

As we have established before, the process will eventually converge, and Euclid algorithm converges to $\gcd$. So, for each prime factor p_j its exponent will converge to

$$D_j = \gcd(d_{1,j}, d_{2,j}, \dots, d_{2026,j})$$

and the only remaining $M > 1$ will be

$$M = \prod_{j=1}^k p_j^{\gcd(d_{1,j}, d_{2,j}, \dots, d_{2026,j})}$$

which depends only on the set of original numbers, not the selection process.